

THE OPEN WINDOW PROBLEM

Simulation Laboratory Design & Development

Draft Pass 0.2

Persistent Interface Exposure • Event/Lineage Closure • Model Credibility • Temporal/Causal Reconciliation
• Root-Conditioned Failure

Document Status	DRAFT PASS 0.2 — controlled working draft; not ONESTAMPED
Date	10 August 2026
Concept origin / project direction	Nathan Lewis II
Documentation preparation	ChatGPT (GPT-5.6 Sol)
Contributor set reviewed	ChatGPT • Claude • Google Gemini • Grok AI • Perplexity AI alternative analysis
Classification	Simulation-only R&D / Net Defense / Worldbuilding Systems Design

SIMULATION-ONLY SAFETY NOTE

No real malware, pathogens, radiation, hazardous chemicals, or physical attack mechanisms are required or authorized by this draft. All “infection,” “bleed,” and “hazard” phenomena are represented by benign synthetic markers and controlled state changes.

0. Draft-Pass Control Record

Draft Pass 0.2 is not a clean-sheet rewrite. It is a controlled consolidation of the v0.1 simulation-lab draft, the origin/doctrine branch, Grok's protocol branch, Gemini's directly supplied scenario/countermeasure ideas, and Perplexity's red-team alternative analysis. Where contributor documents conflict, this pass records the difference and converts disputed claims into testable hypotheses rather than silently choosing a winner.

Change area	0.2 decision	Reason
Document governance	Separate logical artifact identity from file representation (DOCX/PDF/MD).	Prevents duplicate exports from being mistaken for distinct conceptual versions.
Doctrine statement	"Screen = aperture" becomes an Aperture Hypothesis, not an established law.	Allows disconfirmation and avoids confirmation bias.
Architecture	Use two orthogonal maps: topology and authority/privilege.	Root relevance depends on trust/authority, not only layer depth.
Closure	Split BTEC into declared-event closure and observable-effect closure; add provenance lineage.	An event can close while descendants remain active.
Time	Separate clock disagreement from causal-order disagreement.	Timezone/calendar variance is benign; causal inversion is stronger evidence.
Evidence	Require instrumentation-failure independence, deterministic replay, and independent recurrence.	Multiple sensors on one engine are not independent evidence.
Run duration	Use exit-criteria-driven pilot, then 7/14-day persistence campaign only if warranted.	Reconciles short prototype and long campaign models.
Root claims	Adopt R0–R4 reachability/compromise taxonomy.	Prevents "root involvement" from being mislabeled as compromise.
External methods	Use NASA M&S credibility + NIST systems-security/resiliency as primary engineering anchors; structured debrief guidance as secondary.	Better methodological fit than physical simulation-center planning alone.

1. Executive Summary

The Open Window Simulation Laboratory is a simulation-only research program for testing whether an interface's visible presentation state reliably represents its underlying transfer state. The core concern is not that real consumer screens are literal portals. The laboratory models a class of nested environments in which media or interface endpoints may carry state across boundaries, then asks whether apparently closed, idle, black-screen, disconnected, or reset states actually terminate the relationship.

Draft Pass 0.2 hardens the project from a rich worldbuilding concept into a falsifiable experimental architecture. It does this by separating doctrine from measurement; requiring explicit null hypotheses and acceptance criteria; distinguishing topology from privilege; adding model-credibility records; introducing independent instrumentation paths; tracking event descendants as well as event closure; and delaying temporal/root campaigns until simpler interface claims have passed reproducibility gates.

Central 0.2 research question: Can the laboratory distinguish a genuine cross-boundary phenomenon from model error, instrumentation dependence, synchronization artifact, observer contamination, rollback/replay effects, or an incorrect underlying ontology — and can it reproduce that distinction?

2. Basis, Provenance, and Source Roles

The source set contains several synthesis branches rather than a single finalized canon. Draft Pass 0.2 therefore treats them as complementary inputs with explicit provenance.

Source / contributor	Primary retained contribution	0.2 treatment
ChatGPT preliminary v0.1	Interface-state decomposition; H1–H9; test matrix; safe proxies; evidence thresholds; funding tiers; decision gates.	Becomes experimental-method backbone; gaps in replication, metric-to-test mapping, and credibility accounting are filled.
Claude origin + v2.0	Open Window Doctrine naming; asymmetric awareness; convenience/exposure; time-discrepancy and reconciliation/root hypothesis; governance status.	Origin record preserved; root-time mechanism remains REPORTED/UNCONFIRMED until tested.
Grok protocol 0.1	Window Object schema; Layer 0–N stack; infection-as-probe; instrumentation suite; 14-day campaign; operator roles.	Window Object retained; long campaign moved after pilot; operator-blinding proposal gated by oversight.
Google Gemini	Domestic observation-chamber grounding; rapid-cycle media; named containment concepts; physical-bleed vividness.	Countermeasures are experimental conditions, not presumed solutions; harmful examples converted to benign proxies.
Perplexity alternative analysis	Configuration-control critique; metric consolidation; confirmation-bias warning; sensor-agreement illusion; stop/go and governance recommendations.	Governance, metric dictionary, pre-registration, and sensor-independence concerns incorporated; claims of “definitive merges” and healthcare-lab emphasis are moderated.

2.1 External engineering references used in this pass

- NASA-STD-7009B (2024), Standard for Models and Simulations: used as a credibility/requirements/acceptance-criteria anchor for simulation products.
- NASA-HDBK-7009B (2026), implementation guidance for NASA-STD-7009B: used to motivate a model-credibility ledger, evidence traceability, and disciplined model consumption.
- NIST SP 800-160 Vol. 1 Rev. 1: used for systems-security engineering, trustworthiness, and explicit trust/privilege boundaries.
- NIST SP 800-160 Vol. 2 Rev. 1: used for resilience objectives — anticipate, withstand, recover, adapt — and for recovery/containment framing.
- INACSL Healthcare Simulation Standards of Best Practice: used only as a secondary source for structured prebrief/debrief/after-action discipline, not as the primary physical-lab design model.

3. Needs Assessment and Intended Decisions

Perplexity correctly identified that the project had not clearly stated which decisions the lab must support. Draft 0.2 therefore defines the program need before additional scenario growth.

Need	Decision supported	Evidence required
Interface-state ambiguity	Can visible closure be trusted as isolation?	Repeated black-screen/off/closed/disconnected trials with verified positive and negative controls.
Unknown-vector admission	Is recognition required before containment?	Benign unknown marker tested through Interface Airlock vs direct path.

Carrier transformation	Can state survive removal of its original host?	Provenance ledger showing descendant state after original event termination.
Temporal inconsistency	Does timing reveal architecture or merely local convention?	Native clocks + monotonic time + event sequence + causal-order checks.
Observer reliability	Can Layer 0 independently validate Layer 1?	Failure-independent telemetry and deliberate sensor-fault trials.
Root-conditioned failure	Can a lower-layer state influence a privileged dependency?	R0–R4 reachability evidence with privilege-boundary logging; propagation alone insufficient.

4. Controlled Definitions Ledger

Term	Draft 0.2 operational definition
Window / interface	A modeled boundary object capable of carrying one or more state classes between environments.
Presentation state	What an inhabitant sees: active, black, minimized, asleep, error, etc.
Transfer state	Whether any allowed or latent state can cross the boundary, independent of presentation.
Exposure	A target environment or agent receives the marker/state.
Contamination	Received state persists beyond the immediate transfer interval or alters local state.
Persistence	State remains after the initiating interface or carrier is nominally terminated.
Recovery	State returns to a verified baseline and remains there through a defined observation window.
Event closure	The simulation/event system declares an event complete.
Effect closure	All independently observable effects attributable to the event have ceased.
Lineage extinction	No descendant state derived from the event remains active in any monitored carrier.
Root involvement	A privileged/root service observes or reacts to a lower-layer condition.
Root compromise	Persistent or controllable unauthorized influence over root-level state (R4 only).

5. Authoritative Architecture Model for 0.2

5.1 Topology map

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L0 Research / Observer Control Plane
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L1 Inhabited Domestic Environment (four-bedroom baseline)
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|— Window Objects: phone / TV / tablet / laptop / embedded display

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L2 Media / Social / Service Environments

L3 Adjacent or Nested Simulation Domains

LX External / Unknown Domain (optional experimental source)

[Interface Airlock is a boundary function that can be inserted between any two domains; it is not assumed to be a separate universe/layer.]

5.2 Authority / privilege map

Privilege class	Role	Examples
P0	No authority / passive artifact	Rendered content, static marker.
P1	Ordinary local process	App/session/window process.
P2	Shared service	Identity, logging, time sync, storage, message bus.
P3	Orchestration authority	Scheduler, rollback manager, environment controller.
P4	Privileged control	Simulation administration, policy enforcement, cross-layer controls.
P5	Root / ultimate simulation authority	Root state, master orchestration, privileged persistence store.

Important: Topological distance and privilege are independent. A marker can traverse many layers without gaining authority, or make one short hop into a high-value shared service. Root testing must therefore log both location and privilege.

5.3 Window Object minimum schema

Field	Examples / values	Purpose
window_id	W-001	Stable identity across replay.
presentation_state	active / black / minimized / sleep	Separates user-facing state.
power_state	on / standby / off / removed	Avoids “off” ambiguity.
application_state	foreground / background / closed	Front-end process state.
network_state	connected / blocked / isolated / unknown	Communications path.
transfer_state	open / constrained / latent / sealed / unknown	Actual modeled crossing capability.
directionality	A→B / B→A / bidirectional	Flow direction.
airlock_mode	none / observe / translate / normalize / quarantine	Admission-control condition.
clock_domain	CD-#	Native time authority.
privilege_relation	P# service(s) touched	Authority dependencies.
event_log_ref	BTEC record	Declared event path.

provenance_ref	lineage record	Descendant-state path.

6. Model Credibility and Experimental Integrity

Draft 0.2 treats simulation output as evidence requiring credibility assessment, not as self-authenticating truth. Every result must be traceable to the model version, scenario version, configuration, seed, instrumentation path, and acceptance criteria used to produce it.

Credibility ledger field	Required entry
Model version / build	Exact simulation build and dependency set.
Scenario version	Immutable scenario ID and protocol revision.
Initial-state fingerprint	Snapshot/hash or equivalent reproducible state identifier.
Random seed / stochastic configuration	Required for deterministic replay where applicable.
Instrumentation configuration	Sensors, observers, clock sources, processing paths.
Expected / null result	Written before run.
Observed result	Raw observation plus interpreted finding.
Uncertainty / known limitations	Resolution, missing data, shared dependencies, timing error.
Replay result	Same state/seed reproduction status.
Independent recurrence	Equivalent-condition replication status.
Alternative explanations	At least one competing explanation for positive anomalies.
Credibility disposition	Supported / weakened / inconclusive / falsified / blocked.

6.1 Deterministic replay vs independent recurrence

- Deterministic replay asks whether the exact same state, seed, and inputs reproduce the event.
- Independent recurrence asks whether the same class of effect appears under equivalent but independently initialized conditions.
- A one-off anomaly that cannot be replayed or independently reproduced is not promoted to a doctrine-level finding.

7. Instrumentation Independence and Observer Integrity

Multiple telemetry channels count as independent evidence only when their failure modes are meaningfully independent. Different dashboards backed by the same simulation state store are not independent evidence classes.

Channel	Primary observation	Independence requirement
Aperture telemetry	Window state transitions and transfer attempts.	Separate logging path from scenario event generator where possible.
Environment telemetry	Temperature/color/sound/state proxy values.	Independent sampling pipeline or post-run state extraction.
Agent telemetry	Gaze, motion, proximity, behavior.	Independent event timestamps and raw trajectories retained.

Surveillance observer	Third-person/L0 representation.	Must not merely mirror agent perception buffer.
BTEC ledger	Declared event lifecycle.	Immutable event IDs and acknowledgements.
Effect-closure observer	Residual consequence detection.	Not derived solely from BTEC closed flag.
Clock audit	Native/local/master/monotonic/sequence time.	At least two clock authorities or sequence ordering source.
Provenance ledger	Carrier/descendant lineage.	Separate from event-closure semantics.

7.1 Observer Integrity scenario family (new)

ID	Scenario	Pass question
OI-1	Independent Sensor Cross-Check	Do architecturally distinct observers agree on a known-positive marker?
OI-2	Sensor Fault Injection	Does the system detect a deliberately corrupted/lagging telemetry channel?
OI-3	Observer Disagreement	Can the protocol remain neutral when Layer 0 and Layer 1 disagree?
OI-4	Observer-of-Observer	Can a secondary audit path detect primary monitor state changes?
OI-5	Blind-Channel Test	Can a known harmless effect outside the primary detector ontology be recovered by another observer?

8. BTEC 0.2: Event Closure, Effect Closure, and Lineage Extinction

Bi-Directional Total Event Closure remains a central concept, but 0.2 separates three questions that were previously conflated.

Layer	Question	Example failure
BTEC-A: Declared event closure	Did the event system open, propagate, acknowledge, and close the event?	E-123 reports CLOSED while residual state remains.
BTEC-B: Observable effect closure	Did every independently observable consequence cease?	Temperature/color/clock marker persists after E-123 closes.
Provenance / lineage extinction	Did all descendant copies/carriers terminate?	Original device cleared, but agent/message/second device still carries marker.

E-123 (origin event)

- ├ C1 media-frame marker
- ├ C2 device-session state
- ├ C3 environment state
- ├ C4 agent memory/behavior state
- └ C5 second-device state

Event closure is achieved when E-123 closes.

Effect closure is achieved when C1..Cn no longer produce observable effects.
Lineage extinction is achieved only when no active descendant carrier remains.

9. Temporal and Causal Reconciliation Model

Draft 0.2 preserves the key insight that different environments may legitimately use different time systems. A mismatch is therefore evidence only after normalization and competing explanations are tested.

Time field	Meaning
Native wall clock	Local time as represented in that environment.
Calendar model	Day/year/epoch conventions; may differ intentionally.
Simulation time	Model progression time, possibly accelerated or paused.
Monotonic elapsed time	Non-decreasing control-layer interval used for duration comparisons.
Event sequence number	Ordering independent of human-readable time.
Clock authority	Source that controls or corrects local time.
Clock rate	Rate relative to reference domain.
Pause / rollback state	Whether local time stopped, rewound, restored, or forked.
Conversion history	Normalization steps applied before comparison.

Temporal rule: Clock disagreement alone is weak evidence. Causal-order disagreement — where an independently confirmed effect precedes its only permitted cause, or event sequence becomes inconsistent across reconciled domains — is a higher-priority anomaly.

9.1 Temporal evidence levels

Level	Condition	Interpretation
T0	Different local clocks/calendars but consistent conversion.	Normal heterogeneity.
T1	Unexplained clock drift in one domain.	Investigate local clock/processing.
T2	Clock drift correlates with transfer-state anomaly.	Correlated anomaly; not root evidence.
T3	Multiple independent domains show coordinated correction/reset.	Shared service or orchestration hypothesis strengthened.
T4	Causal-order or sequence contradiction survives replay and independent recurrence.	High-priority reconciliation defect.
T5	Lower-layer time state measurably influences privileged synchronization/orchestration behavior.	Candidate root-conditioned influence; map with R taxonomy.

10. Pre-Registered Hypotheses and Falsification Criteria

Hypothesis	Null / falsification condition	Required evidence
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H1 Presentation $\neq$ transfer	Repeated inactive/black states show no transfer while positive controls succeed.	Window telemetry + independent effect observer.
H2 Lower-layer persistence	Closed app never leaves transfer/descendant state across replications.	Process/window state + provenance.
H3 Multi-window exposure	Exposure probability is unchanged across interface-count conditions.	Matched runs with controlled concurrency.
H4 Carrier switching	No descendant marker survives original-carrier termination.	Provenance lineage and delayed observation.
H5 Awareness effect	Aware vs unaware agents show no meaningful response/containment difference.	Behavioral metrics with matched scenarios.
H6 Temporal mapping	Time mismatch does not correlate with any independent layer/service relationship.	Clock audit + service dependency data.
H7 Band-Aid closure	Visible/administrative closure always equals effect closure and lineage extinction.	BTEC-A/B + provenance.
H8 Root traceability	No lower-layer state produces independently verifiable privileged-system influence.	Privilege map + R classification.
H9 Interface Airlock	Airlock condition does not reduce marker propagation vs matched direct condition.	Paired trials with unknown-marker class.

10.1 Multi-vector corroboration scale

Grade	Evidence pattern	Disposition
C0	Single isolated anomaly.	Log; no infection claim.
C1	Two correlated observations but shared architecture possible.	Investigate dependence.
C2	Cross-modal observations with at least one failure-independent path.	Candidate cross-boundary event.
C3	Cross-layer event reproduced deterministically and independently.	Supported phenomenon for tested conditions.
C4	Nominal containment followed by verified residual/descendant state.	Closure failure supported.
C5	Privileged dependency influence verified under R2+ criteria.	Root-conditioned effect; not necessarily compromise.

11. Draft Pass 0.2 Minimum Viable Protocol (MVP)

The MVP deliberately postpones carrier-switch stress, extended time-divergence, and root-reachability campaigns. Its first task is to prove that the instrumentation, event accounting, and falsification logic work.

Run	Purpose	Core condition	Required exit criterion
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P0	Instrumentation calibration	Known-negative baseline + known-positive benign marker.	Positive detected; negative remains negative; timestamps/IDs reconcile.
P1	Presentation vs transfer	Active / black / minimized / closed / disconnected states.	At least 3 replications per selected state; falsification recorded if no transfer.
P2	Total closure	Trigger marker, close interface administratively, observe residual window.	BTEC-A/B and provenance reconcile; false-success condition detectable.
P3	Carrier-neutral content marker	OW-MARKER-01 transferred via harmless color/temp/checksum/time token.	Transfer mechanics observed without assuming biological/glitch ontology.
P4	Observer integrity	Fault one telemetry path while independent observer remains clean.	System identifies disagreement; no automatic majority-vote truth assumption.
P5	Replay / recurrence	Replay one positive and one null condition; then independent fresh run.	Credibility ledger completed with replay and recurrence outcome.

11.1 Pilot duration is exit-criteria-driven

The prior drafts proposed both a roughly 12-hour/two-day prototype and a 14-day campaign. Draft 0.2 reconciles these by making the pilot criteria-driven rather than calendar-driven. A nominal planning envelope is 8–16 active hours plus one unattended persistence window, but the pilot does not advance until the gates below pass. A 7-day controlled campaign and 14-day persistence campaign are authorized only after the MVP produces credible, repeatable data.

Gate	Advance only if...
G0 Configuration	Artifact IDs, model version, scenario version, seed/configuration are frozen.
G1 Instrumentation	Known-positive/negative tests and sensor-fault tests pass.
G2 Closure accounting	BTEC-A/B and lineage records reconcile on control events.
G3 Replay	At least one positive or null result replays deterministically as expected.
G4 Independent recurrence	Equivalent fresh run produces a consistent result class.
G5 Recovery	Rollback/reset returns system to verified baseline without hidden descendant state.
G6 Review	After-action panel classifies findings and approves expansion.

12. Campaign Expansion After MVP

Campaign	Nominal envelope	Adds	Purpose
C1 Controlled campaign	~7 simulated days	Aware/unaware; convenience gradient; multi-window; carrier switching.	Generalize interface and human-factor findings.

C2 Persistence campaign	~14 simulated days	Delayed activation; residual accumulation; intermittent faults; repeated closure/recovery.	Test slow-burn effects and false recovery.
C3 Temporal campaign	Criteria-driven	Independent clock domains; pause/resume; rollback; rate changes; causal ordering.	Test architecture inference without overreading benign clock differences.
C4 Privilege/root campaign	Highest-security posture	Shared services, synchronization, orchestration, privilege boundaries.	Classify R0–R4 influence; R4 only may be called root compromise.

13. Conditional Failure at the Root Level — 0.2 Annex

The Open Window model connects to Conditional Failure at the Root Level when a lower-layer condition causes an upstream privileged component to violate an invariant it is supposed to maintain. The window is therefore a possible tracer or dependency entry point, not automatically a root exploit.

Grade	Definition	Allowed claim
R0	No upstream reach beyond originating domain.	Local-only phenomenon.
R1	Privileged/upstream component observes or responds.	Root/upstream involvement.
R2	Lower-layer state measurably changes privileged behavior.	Upstream influence.
R3	State crosses a privilege/trust boundary contrary to policy.	Privilege-boundary violation / candidate compromise path.
R4	Persistent or controllable unauthorized influence over root-level state.	Root compromise supported for tested conditions.

13.1 Temporal reconciliation root hypothesis (still unconfirmed)

The Claude/Perplexity branch proposes that contradictory timestamps might interact with a reconciliation service that touches many layers. Draft 0.2 preserves this as a candidate mechanism but does not assume that the service trusts window-reported time, escalates privileges, or exposes root state. The experiment must first determine the clock authority, verification path, and privilege of any reconciliation component.

1. Map clock authorities and identify which services can correct local time.
2. Introduce benign, bounded timestamp/sequence disagreement within allowed lab parameters.
3. Observe whether the discrepancy remains local, is normalized, is rejected, or causes a privileged response.
4. Repeat with independent sequence-order evidence to distinguish clock formatting from causal contradiction.
5. Classify outcome R0–R4; do not infer R4 from resource allocation, logging, or root observation alone.

14. Safety, Ethics, and Stop Conditions

Even inside a fictional/simulated program, the lab requires explicit stop-loss rules. The purpose is disciplined containment testing, not uncontrolled escalation.

Trigger	Immediate action	Resume condition
Unexpected propagation outside authorized domains	Freeze new injections; isolate test branch; snapshot state.	Boundary restored; provenance reconciled; review approves restart.

Instrumentation integrity loss	Pause interpretive claims; preserve raw data; switch to independent observer if available.	Known-positive/negative sensor checks pass.
Unbounded event/lineage growth	Terminate source condition; enforce hard resource cap.	Lineage catalog complete and baseline restored.
Rollback/recovery fails	Stop campaign; do not advance to temporal/root testing.	Clean recovery demonstrated twice.
Operator awareness manipulation proposed	Require pre-authorized oversight plan, scope, recovery cue, and independent observer.	Governance approval and rehearsed restoration path.
R3 candidate privilege crossing	Escalate to root-campaign control posture; no further privilege testing automatically.	Independent review confirms scope and containment.

Operator blinding/partial unawareness is NOT authorized by Draft 0.2 by default. It is an optional later human-factors protocol requiring explicit oversight, bounded deception, predefined restoration, and an independent safety observer.

15. Unified Metrics Dictionary

Metric	Definition	Maps prior terminology
TTE	Time to exposure.	Retained.
TTD	Time to detection.	Retained / aperture residual detection timing.
TTA	Time to attribution.	Retained.
TTC	Time to containment.	Retained / containment success.
FCR	False closure rate: declared closed while effect/lineage remains.	Expands prior false-closure concept.
CTC	Carrier transition count.	Retained / carrier-switch measure.
PD / PB	Propagation depth / breadth.	Retained.
ECR	Effect closure rate.	Formalizes BTEC-B.
LER	Lineage extinction rate.	New: descendant-carrier eradication.
IIR	Instrumentation independence ratio: evidence channels judged failure-independent / total channels used.	New response to sensor-agreement illusion.
RRR	Replay/recurrence reliability.	New model-credibility metric.
TCR	Temporal causal-reconciliation anomaly rate.	Separates causal-order issues from clock-format mismatch.
R-grade	R0–R4 highest verified privilege influence.	Replaces ambiguous root reachability claims.
CCR	Credibility completeness rate: required ledger fields completed.	New NASA-style credibility accounting proxy.

16. Resource and Funding Framework

Prior drafts used two different planning languages: dollar tiers and person-hours. Draft 0.2 retains both but ties them to scope rather than treating them as competing estimates. Dollar figures remain placeholders until platform, staffing rates, and compute environment are defined.

Tier	Scope	Labor planning	Illustrative budget envelope*
Lean MVP	P0–P5 only; existing platform; minimal custom visualization.	~40–80 skilled hours.	\$5k–\$15k
Standard controlled	MVP + 7-day campaign; more telemetry and analysis.	~80–160 skilled hours.	\$25k–\$60k
Extended persistence	14-day campaign; shift coverage; higher storage/compute.	~160–300+ skilled hours.	\$60k–\$120k
High-fidelity/root	Dedicated environment, deeper privilege mapping, extensive replay/analysis.	Project-specific.	\$100k–\$250k+

*Planning envelopes inherited/refined from earlier drafts; not procurement quotes. Re-estimate after platform and staffing assumptions are fixed.

16.1 Minimum roles

- Simulation operator / run controller.
- Instrumentation and telemetry analyst.
- Scenario designer / red-team lead for benign marker injection.
- Model-credibility / V&V reviewer independent from scenario author when feasible.
- After-action / documentation lead.
- Optional independent safety observer for unattended, observer-integrity, or later operator-blinding trials.

17. Artifact Governance and Multi-AI Review Provenance

Draft Pass 0.2 distinguishes a logical artifact from its file formats. A PDF and DOCX carrying identical content are representations, not separate versions.

Artifact ID	Purpose	Status rule
OWL-ORIGIN-001	Original brainstorm / lineage record.	Never overwritten; archival.
OWL-DOCTRINE-001	Conceptual doctrine and assumptions.	May be revised independently of protocol.
OWL-MODEL-001	Authoritative topology, privilege map, Window Object schema, metric dictionary.	Single source for architecture definitions.
OWL-PROTOCOL-001	Executable simulation protocol and run gates.	This Draft Pass 0.2 is the precursor.
OWL-ROOT-ANNEX-001	Root-conditioned failure / privilege testing annex.	Remains gated until MVP data is credible.
OWL-AAR-###	After-action/result records.	Immutable after approval; references exact model/scenario version.

17.1 Required AI contribution provenance header

Contributor:

Input documents seen:

Prior AI responses seen:

Role: independent analysis / red team / synthesis / documentation

New proposals:

Inherited proposals:

Disagreements:

Confidence / uncertainty:

Date / model or system identifier:

This prevents later convergence from being misread as independent validation. Cross-informed agreement is still valuable, but it must be labeled as dependent review rather than independent replication.

18. Prebrief, Debrief, and After-Action Structure

Stage	Required outputs
Prebrief	Run objective; hypotheses/nulls; known risks; stop criteria; role assignments; instrumentation status; version IDs.
Hot debrief	What happened; what surprised operators; instrumentation disagreements; deviations from protocol.
Technical reconciliation	BTEC-A/B; provenance; time/sequence; replay; sensor independence; uncertainty.
Red-team challenge	At least two alternative explanations for every major positive finding; identify confirmation paths.
Disposition review	Supported / weakened / inconclusive / falsified / blocked; authorize or deny next campaign.
Documentation update	Change log; metric dictionary updates; artifact version; no silent canonization of unverified mechanisms.

19. Consolidated Action Plan for Pass 0.2 → Executable v0.3

- Approve the 0.2 glossary, topology map, privilege map, Window Object schema, and unified metric dictionary as the working model baseline.
- Assign artifact IDs and create a change log that treats DOCX/PDF/MD as representations of the same logical artifact where content matches.
- Write the six MVP scenario cards (P0–P5) with exact inputs, nulls, positive/negative controls, replications, metrics, and stop conditions.
- Engineer or simulate at least one failure-independent observation path and a deliberate telemetry-fault test.
- Implement BTEC-A, BTEC-B, and lineage/provenance records with immutable event/carrier IDs.
- Implement the expanded time model: native clock, simulation time, monotonic time, sequence number, authority, pause/rollback, conversion history.
- Run the MVP until G0–G6 exit criteria pass; do not advance merely because a calendar window has elapsed.
- Use MVP results to choose whether the next campaign is 7-day controlled, 14-day persistence, temporal, or blocked for redesign.
- Only after credible temporal/service mapping should the Root Annex execute R0–R4 privilege tests.
- Circulate the resulting v0.3 draft and AAR to the multi-AI team with provenance headers so independent vs cross-informed analysis remains visible.

20. Deferred but Required Future Elements

- Formal statistical power/sample-size plan once expected effect sizes are known. Draft 0.2 uses minimum replications for engineering validation, not inferential-statistics claims.
- A machine-readable event/provenance schema and query format for automated reconciliation.
- Formal threat model identifying adversary capabilities, objectives, and assumed knowledge for the Net Defense exercise.
- Performance/load testing for hundreds or thousands of simultaneous windows after basic mechanics are validated.
- Model drift detection across software revisions so an old “clean” baseline is not silently reused after architecture changes.
- Cost model tied to actual compute platform, storage rates, staffing rates, and fidelity requirements.
- A canon/governance decision on ONESTAMP authority remains organizational and is not assigned unilaterally by this draft.

21. References and Working Source Set

21.1 Internal working documents

- Open_Window_Simulation_Lab_Preliminary_Draft_v0.1 (ChatGPT; 9 Aug 2026).
- OAL-Open-Window-Infection-Scenario-v1.0 (Claude structuring layer; 9 Aug 2026).
- OAL-Open-Window-Doctrine-Unified-Draft-v2.0 (Claude synthesis; 9 Aug 2026).
- Open_Window_Doctrine_Simulation_Lab_Protocol_Draft_0.1 (Grok synthesis; 9 Aug 2026).
- Google Gemini direct response supplied in multi-AI pass-through (9–10 Aug 2026).
- Open Window Simulation Lab — Consolidated Alternative Analysis & Action Plan (Perplexity AI; 10 Aug 2026).

21.2 External methodological references

- NASA. NASA-STD-7009B, Standard for Models and Simulations. Approved 5 March 2024. <https://standards.nasa.gov/standard/NASA/NASA-STD-7009>
- NASA. NASA-HDBK-7009B, NASA Handbook for Models and Simulations: An Implementation Guide for NASA-STD-7009B. Approved 3 February 2026. <https://standards.nasa.gov/standard/NASA/NASA-HDBK-7009>
- NIST. SP 800-160 Vol. 1 Rev. 1, Engineering Trustworthy Secure Systems. <https://csrc.nist.gov/pubs/sp/800/160/v1/r1/final>
- NIST. SP 800-160 Vol. 2 Rev. 1, Developing Cyber-Resilient Systems: A Systems Security Engineering Approach. <https://csrc.nist.gov/pubs/sp/800/160/v2/r1/final>
- INACSL. Healthcare Simulation Standards of Best Practice. <https://www.inacsl.org/healthcare-simulation-standards-of-best-practice->

END-OF-PASS STATUS: Draft Pass 0.2 establishes the hardened experimental architecture. It does not ONESTAMP the Open Window Doctrine, prove the Aperture Hypothesis, establish a biological/physical transfer mechanism, or claim root compromise. The next executable deliverable is an MVP scenario-card set and instrumentation schema.